Ayesha Firdaus

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PROFESSIONAL SUMMARY

Data Analyst with a biopharma operations background and hands on experience in SQL, Python, Power BI, and ML. Built four end to end projects across fraud detection, dynamic pricing, demand forecasting, and inventory optimization, each with measurable business outcomes

SKILLS

Languages: SQL, Python (Pandas, NumPy, Scikit-learn, Matplotlib)

BI & Reporting: Power BI, Excel, Looker Studio, Metabase

ML & Modeling: LightGBM, XGBoost, Regression, Classification, Time Series Forecasting, Hypothesis Testing, Feature Engineering

Data & Tools: EDA, Data Cleaning, ETL/ELT, Data Modeling, Git, Google BigQuery

Gen AI & LLM: OpenAI/Anthropic API, Prompt Engineering, Structured Outputs, Function Calling, Agent Basics (OpenAI Agents SDK, MCP)

WORK EXPERIENCE

Intern - MSAT — 09/2024 – 05/2025 | Bengaluru, India
Syngene International

- Analysed batch manufacturing data across multiple client projects — tracked parameter distributions, identified outliers, and escalated deviations to manufacturing teams; corrective actions on one key protein purification project improved batch purity from ~90% to 98%
- Built Excel-based tracking and analysis workflows for parameter monitoring across batches; supported root cause investigations for out-of-range values and contributed to process deviation reports submitted to stakeholders
- Prepared base documents and TTDs covering raw materials, consumables, quantities, and vendor specifications; coordinated with QC teams to validate inputs and close documentation loops across active projects
- Supported scale-up calculations for buffer preparations and raw material quantities during lab-to-manufacturing transitions; produced calculation sheets used by manufacturing teams for procurement and process planning
- Worked across projects for clients in pharmaceutical and nutrition sectors (including BMS and Zoetis) in a downstream processing environment; handled documentation, cross-functional coordination, and analytical requests in parallel

PROJECTS

Banking Fraud Detection & Transaction Risk Analytics — 03/2026 – 04/2026
SQL Server, Python, XGBoost, Power BI

- Designed a four layer fraud detection system: SQL analytics, behavioural risk scoring, XGBoost (ROC AUC 0.844), and Power BI dashboards across 44K+ transactions; identified $380M in fraud exposure in synthetic dataset
- Custom risk engine stratified transactions into four bands; Critical Risk segment isolated at 60%+ fraud rate, flagging 665 high-priority customers for investigation
- Key fraud signals — crypto merchants (40.1% fraud rate), 1AM peak transactions (23.4% fraud rate), and fraud occurring ~6x farther from customer home locations, surfaced via 4-page executive dashboard

Customer Channel Affinity Prediction System — 04/2026 – 05/2026
Python, LightGBM, Streamlit

- Developed a multiclass ML model using 140+ engineered behavioural features across 9 data sources to predict future customer channel affinity
- Implemented leakage safe temporal validation; LightGBM achieved 33.5% top-1 / 70.4% top-3 accuracy (2.3x lift over 14.3% random baseline)
- Deployed full pipeline as a live Streamlit app with customer-level channel recommendations, probability rankings, and budget allocation simulator

Dynamic Pricing & Demand Forecasting System — 03/2026 – 05/2026
Python, LightGBM, Streamlit

- Built an end to end ML pipeline forecasting hourly demand and computing revenue optimal dynamic prices
- Engineered 23 features on 8,737 hourly records using lag windows, DSR, and event flags
- Achieved 3.1% MAPE vs 16.7% naive baseline, delivering +12.7% revenue uplift (₹1.45M annualised) with 83.7% demand retention

Supply Chain Inventory Optimization & Demand Forecasting — 04/2026 – 05/2026
SQL Server, Python, Power BI

- Built an end to end analytics pipeline diagnosing inventory inefficiencies across 6 supply chain tables; engineered reorder points, safety stock, and turnover KPIs in SQL; built 30-day rolling demand forecasts in Python
- Identified ~95% of products were overstocked, total inventory held at ~23x forecasted monthly demand; surfaced via a 3 page executive Power BI dashboard with product level drill down and supplier delay analysis
- Flagged Electronics as the lowest turnover, highest priority category for inventory reduction; quantified a potential reduction of ~196 units per product against current avg stock of ~221

EDUCATION

Vellore Institute of Technology — 09/2020 – 08/2025 | Vellore, India
MSc. Integrated Biotechnology
CGPA - 8.31/10
Relevant coursework: Statistics, Data Analysis, Research Methodology

AnalytixLabs & E&ICT Academy, IIT Guwahati — 08/2025 – Present | Bengaluru
Advanced Certification in Data Science (ACDS)